LESSON PLAN · WEB DEVELOPMENT (GR 6–8) BEGINNER: WEB CREATORS · LEVEL 1

Lesson 5: Classes, flexbox and cards

Module 2 | 60 minutes | classes, flexbox, cards | Grades 6–11 · ages 11–16

Objective: Use class selectors and flexbox to lay out a row of styled cards with gaps between them.

Success criteria

- I can add a class and style it with a dot.
- I can group elements in a div.
- I can line up cards with display: flex and gap.
- I can style a box to look like a card.

Key vocabulary

- **class**: A group name given to elements, like class="card".
- **class selector**: A CSS selector with a dot, like .card.
- **div**: A plain box used to group elements.
- **flexbox**: A CSS layout that lines up children in a row or column.
- **gap**: The space between flex items.
- **parent / child**: A box that holds other boxes / a box inside another.

You will need

- Projector with Kitabash (lesson video, Web Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: l05-fruit-cards.html, l05-teacher-demo.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Flex Line-up**	Play Flex Line-up with five volunteer “cards”. Finish with the “.red!” class round.
5 min	**Hook and video**	Show the hook. Poll: “Name an app that uses cards.” Play the video; pause at “Spot the bug” and let learners find both dots.
10 min	**Live demo and build-along**	Run the demo as written, including the deliberate mistake. Everyone builds the row of fruit cards.
8 min	**Challenge and games**	Web Lab: line up the cards, then Predict the class, the match game and Order it. Quick competition: first correct gallery order wins.
17 min	**Project in pairs**	Breakout rooms: pairs agree on a Top 5 topic and build the gallery. Driver types, navigator checks dots and brackets; swap every 5 minutes.

5 min	**Probe**	New situation: cards still stacking. Ask two learners to explain parent and child in their own words.
6 min	**Show-and-tell**	Gallery tour: four learners share their Top 5; class reacts with emoji for their own number 1 pick.
4 min	**Wrap-up and exit ticket**	Thumbs check on each success criterion. Exit ticket: the .row flex rule.

Unplugged warm-up

Flex Line-up (5 min game): five volunteers are “cards” and the teacher is the parent box. Call “display: flex!” and they stand in a row; “gap: big!” and they spread out; “flex-wrap!” and the row splits onto two lines when it hits the wall. Then call “.red!” and only learners wearing red (the class) wave.

Teaching the idea

A class is a group name we add in HTML with class="..." and select in CSS with a dot. A div is a plain box for grouping. Boxes normally stack, but display: flex on a parent puts its children in a row, and gap adds space between them. flex-wrap: wrap moves items to a new line when space runs out. Cards are just divs with padding, a border and rounded corners.

Analogy: Flexbox is like a cricket team lining up for the national anthem: the captain (parent) says “line up!” and the players (children) stand side by side with a gap between each.

Live demo: 1) Type three boxes with emoji fruits and show they stack on top of each other. 2) Add class="card" to each div and write a .card rule with padding, a border and border-radius. Stress the dot in front of card in CSS, and no dot in the HTML. 3) Wrap the three cards in . Add .row { display: flex; } and watch them jump into a line. 4) Add gap: 16px; to .row and show the space appear between the cards. 5) Add two more cards and flex-wrap: wrap; then shrink the preview to show cards moving to a new line. 6) Move display: flex onto .card by mistake and show it does not line up the cards — flex goes on the parent.

Questions to ask

Ask	Listen for
How do you select class="card" in CSS?	With a dot: .card
Where does display: flex go?	On the parent box that holds the items.
What does gap do?	Adds space between the flex items.
Why use a class instead of styling all divs?	So you can style only some elements, not every div on the page.
What makes a box look like a card?	Padding, a border, rounded corners and a background colour.

Differentiation

Support
Give the build-along code to start the project and ask learners to change only the words and colours. A dot-or-no-dot card: HTML no dot, CSS dot.

Stretch
Use justify-content: center, give cards a hover effect with .card:hover { transform: scale(1.05); }, and add box-shadow.

Common misconceptions

Misconception	How to fix it
Writing class=".card" in HTML	No dot in HTML. The dot is only in CSS.
Putting display: flex on every card	Flex goes on the parent; its children line up.
Using margin on every card instead of gap	gap on the parent adds even spacing in one line.
A class can only be used once	Many elements can share a class; that is the point of it.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write the CSS that puts the children of .row side by side with 20px between them.

Homework
Make a second Top 5 gallery on a new topic, or add flex-wrap and a hover effect to today’s gallery. Show it to someone at home and ask which card they would put first.

Answer key — Kitabash activities

1. Web Lab: line up the cards (perform · web lab)

Goals checked: Uses flexbox · Adds a gap · Styles the row with a class selector

One solution:

```
<style>
  .row {
    display: flex;
    gap: 16px;
  }
  .card {
    border: 2px solid teal;
    padding: 12px;
  }
</style>

<div class="row">
  <div class="card">🏏 Cricket</div>
  <div class="card">🏸 Badminton</div>
  <div class="card">♟️ Chess</div>
</div>
```

2. Predict the class (perform · code output)

3. Match each piece of code to what it does (perform · matching)

class="card" → Gives an element a group name in HTML · .card { } → Styles every element with that class · → A plain box for grouping things · display: flex; → Lines up the children side by side · gap: 16px; → Adds space between flex items · border-radius: 12px; → Rounds the corners of a box

4. Order it: build a gallery (perform · ordering)

Order: Type a div for each card with class="card" → Style .card with padding, border and border-radius → Wrap all the cards in a div with class="gallery" → Give .gallery display: flex and a gap → Add flex-wrap: wrap so cards move down on small screens

5. Project: my Top 5 card gallery (perform · web lab)

Goals checked: Has five cards with headings · Uses classes · Uses flexbox with a gap · Styles the cards as boxes · Has a description on each card

Example page:

```
<!-- My top 5 Indian festivals -->
<style>
  body {
    font-family: Verdana, sans-serif;
    background-color: #fff8f0;
    color: #2c3e50;
  }
  h1 {
    text-align: center;
    color: #c0392b;
  }
  .gallery {
    display: flex;
    flex-wrap: wrap;
    gap: 16px;
    justify-content: center;
  }
  .card {
    width: 180px;
    background-color: white;
    border: 2px solid #e67e22;
    border-radius: 12px;
    padding: 14px;
    text-align: center;
  }
  .rank {
    font-size: 28px;
    color: #e67e22;
  }
</style>

<h1>My Top 5 Festivals</h1>
<div class="gallery">
  <div class="card">
    <p class="rank">#1</p>
    <h3>🪔 Diwali</h3>
    <p>Lamps, rangoli and sweets with the whole family.</p>
  </div>
  <div class="card">
    <p class="rank">#2</p>
    <h3>🎨 Holi</h3>
    <p>Colours everywhere and gujiya to eat.</p>
  </div>
  <div class="card">
    <p class="rank">#3</p>
    <h3>🪁 Makar Sankranti</h3>
    <p>Flying kites from the rooftop all day.</p>
  </div>
  <div class="card">
    <p class="rank">#4</p>
    <h3>🌙 Eid</h3>
```

```
    <p>New clothes, sheer khurma and visiting friends.</p>
  </div>
  <div class="card">
    <p class="rank">#5</p>
    <h3>🎄 Christmas</h3>
    <p>Plum cake, carols and a star on the door.</p>
  </div>
</div>
```

6. New situation: cards still stacking (probe · multiple choice)

Answer: Put display: flex on the parent box that holds the cards

Answer key — worksheet

1. **Key words:** class = A group name given to elements, like class="card". · class selector = A CSS selector with a dot, like .card. · div = A plain box used to group elements. · flexbox = A CSS layout that lines up children in a row or column.
2. **Plan your page:** Goals: Uses flexbox; Adds a gap; Styles the row with a class selector.
3. **Predict the class:** A. Riya and Meera
4. **Match each piece of code to what it does:** class="card" → Gives an element a group name in HTML · .card { } → Styles every element with that class · → A plain box for grouping things · display: flex; → Lines up the children side by side · gap: 16px; → Adds space between flex items · border-radius: 12px; → Rounds the corners of a box
5. **Order it: build a gallery:** 1 Type a div for each card with class="card" · 2 Style .card with padding, border and border-radius · 3 Wrap all the cards in a div with class="gallery" · 4 Give .gallery display: flex and a gap · 5 Add flex-wrap: wrap so cards move down on small screens

Kitabash original curriculum.